GEK-83833

DRIVE SYSTEM INSTRUCTIONS

POWER SYSTEM STABILIZER 3S7932LA202G12

NOTE: Includes frequency transducer with input 50 Hz bandpass filter.

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

POWER SYSTEM STABILIZER

3S7932LA202G12

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NOTICE

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NOTE	Commands attention to an essential operating or maintenance procedure, condition or statement which must be highlighted.

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator frequency, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a signal from generator terminal voltage (3 - phase). This signal, which is frequency modulated, is passed through a bandpass filter to remove excessive power system noise and a frequency transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer, signal conditioning, limiting and output circuitry. Figure 6 is a typical functional elementary diagram while Figure 7 is composed of the printed circuit board elementary diagrams.

The Power System Stabilizer will apply the following transfer function to modify the output of the speed transducer for insertion into the voltage regulator:

$$G(s) = \frac{K(T_1S)\ (1 + T_2S)\ (1 + T_4S)}{(1 + T_1S)\ (1 + T_3S)\ (1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 10% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges:

K	=	0.1	to 100	
T_1	=	0.1	to 50	seconds
T_2	=	0.2	to 1.5	seconds
T_3	=	0.02	to 0.1	seconds
T_4	=	0.2	to 1.5	seconds
T_5	=	0.02	to 0.1	seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to a least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

ac	Alternating Current
Amp	Amperes
dB	Decibels
dc	Direct Current
FET	Field Effect Transistor
GRD	Ground
Hz	Hertz
IC	Integrated Circuit
ma	Milliamp

ABBREVIATIONS (Continued)

mv	Millivolt
PP	Peak-to-Peak
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
VDC	Direct Current Voltage

DESCRIPTION

The Power System Stabilizer consists of:

3S7932LA202G12	Panel

The Power System Stabilizer is normally mounted in either the regulator cubicle or exciter cubicle and contains one each of the following:

Number	Name
44C372627G02	50 Hz Bandpass Filter Board
44C331856G02	Frequency Transducer Board
44C331857G03	Signal Conditioning Board
44C331859G01	Washout and Output Board
44C331863G02	Relay Board
44C331861G01	Ceiling Sensing Board
44C331862G01	Power Supply Board
3S7932MA321G11	Internal Frequency Panel
3S7932MA321G9	± 32 Volt Power Supply Panel
3S7932MA321G13	24 Volt Power Supply Panel

INSTALLATION

LOCATION AND CONNECTIONS

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.

INITIAL CHECKS

1. Wire check all external connections to the PSS. Be sure that wire size and type are adequate for the expected signal levels. Refer to the connection diagram furnished with the excitation equipment.

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* CAUTION *
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DO NOT HI-POT OR MEGGER ANY INTERNAL CIRCUIT AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR EXTERNAL CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Ensure that all printed circuit boards are fully inserted.

3. Ensure that all shielded cables have properly terminated shields.

OPERATION

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* CAUTION *
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MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

OPERATION (Continued)

INITIAL OPERATION

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* CAUTION *
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THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE MANUAL MODE FOR ALL TESTS THROUGH PARAGRAPH 10 (EXCEPT PARAGRAPH 2 AS NOTED). FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1. Power Supply Tests: Turn on ± 15 VDC power using the toggle switch mounted on the faceplate of the Power Supply Board (LOC 1111). Check that power supply voltages are correct.

   Adjustment of the ± 15 VDC supply can be made with board mounted rheostats. (Refer to theory of operation section.)

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* CAUTION *
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ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. ENSURE THAT THE REGULATOR IS IN THE MANUAL MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

2. Relay Board Tests: If the generator is in the automatic mode of operation, transfer to manual. Pull the Relay board (LOC 1112). The Relay board is designed to interface with many excitation systems. Board mounted shorting clips and rheostats are used to adapt the Relay board to the particular excitation system. (Refer to the drawings furnished with the equipment for the settings fo F1S, F2S, F3S, F4S, F1P and F2P.) Check settings and readjust if necessary. Insert Relay board. Pull Washout and Output board (LOC 1108). Put shorting clip C5S on terminals 1 and 2 (not GRD). Insert Extender board in LOC 1108.

OPERATION (Continued)

INITIAL OPERATION (Continued)

2. Relay Board Tests (continued):

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* CAUTION *
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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN MANUAL MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11 and 12 to pins 3 and 4 of the Extender board. This signal can be a dc signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" LED or pushbutton being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" circuit will not stay energized, then press the reset pushbutton mounted on the faceplate of the Ceiling Sensing board. If the "ON" circuit will still not stay energized, then check that the permissive contact is closed, usually 52G line breaker (see sheet 4 of Figure 6). If the unit is off line, then this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" pushbutton or LED should be lighted. Remove the Extender board. Insert the Washout and Output board in LOC 1108. On the Ceiling Sensing board (LOC 1110), put the delay dial on zero, the positive limit dial on 500 (5 turns) and the negative limit dial on 500. Press the reset pushbutton on the Ceiling Sensing board.

Adjust the gain for 1.0 by turning the dial marked "Gain K" on the Washout and Output board to zero. Connect a variable source of dc voltage to "IN 1" of the Washout and Output board and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to $\approx$ 6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" pushbutton to be lighted. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the "ON" position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service (with the generator control in manual) by pressing the "ON" pushbutton. The PSS should latch in the "ON" position and the "ON" LED or pushbutton

OPERATION (Continued)

INITIAL OPERATION (Continued)

2. Relay Board Tests (continued):

should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" pushbutton or LED should be lighted. Some installations may have customer controls for the PSS mounted in the control room. Check that any external customer controls can be used to insert and remove the PSS. Leave the PSS in the "OFF" position.

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* CAUTION *
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THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AUTOMATIC MODE. ENSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. ENSURE THAT THE FREQUENCY TRANSDUCER BOARD IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output will cause a one percent change in generator open circuit terminal voltage. With the PSS off, a small step of voltage, $\approx$ 1 VDC, should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off line. The test may be satisfactorily performed on line if the reactive current compensator (if any) is bypassed and if the voltage regulator of any other generator units paralleled at terminal voltage is temporarily turned to manual control. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt $\pm$ 0.1. The minus sign, terminal voltage decreases for a positive input voltage, arises because the signal is inserted into IN1, which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain if necessary.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in manual, or test if a test position is available on the control switch, pull the Frequency Transducer (LOC 1101). Place the PSS in service using the pushbuttons on the Relay board. Note whether the ac regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P on the Relay board to minimize the change in ac regulator output voltage

OPERATION (Continued)

INITIAL OPERATION (Continued)

2. Relay Board Tests (continued):

when the PSS is inserted. Remove the voltage at IN1 and remove the PSS from service.

3. Internal Frequency Panel Tests: Check out the Internal Frequency panel. This panel is mounted below the printed circuit board rack and contains voltage transformers which accept three phase voltage (115VAC) and provide an output voltage proportional to phase B line-to-neutral voltage. Check that the three phase voltage having the correct amplitude and phase sequence is connected to L25, L26 and L27 (ØA, ØB and ØC, respectively). With 115 VAC on the primary of L1T and L2T and the isolating switches closed, the voltage measured from LA2-A to LA2-D should be 3.5 VAC $\pm$ 0.5 VAC. Secondary taps are provided on L1T and L2T to compensate for a $\pm$ 5 volt variation in primary voltage. For minimum noise, a shielded twisted cable should be used from the voltage source to terminals L25, L26 and L27. Ground the shield at the PSS panel only. The Internal Frequency panel can be used to generate a voltage having so-called "internal frequency" by vectorially adding a phase shifted current signal to the voltage signal. This is done using a reactor and a current signal proportional to phase B generator line current. The voltage developed is then: $V = V_T + jIX$. The value of the reactance is normally chosen to be equal to X_q in per unit.

If it is desired to use the "internal frequency" feature, a shorting jumper from LA2-D to LA2-B must be removed and the shorting switch in the primary of LIX closed. With rated voltage on the generator and supplying unity power factor load current, a dual trace scope (or 2 channel brush recorder) can be used to determine if the voltage and current signal have the proper phasing. For a unity power factor load, the voltage from LA2-D to LA2-B should lead the voltage from LA2-A to LA2-D by 90 degrees. For underexcited operation (leading power factor), the voltage will lead by more than 90 degrees and for overexcited operation (lagging) the voltage will lead by less than 90 degrees. If a two channel oscilloscope is used, connect the common of the oscilloscope to LA2-D with one probe on LA2-B and the second probe on LA2-A. This will cause an inversion of the voltage signal so use the inversion feature of the oscilloscope to invert channel two. The signal on channel one, which is proportional to line current, will lead the voltage by 90 degrees at unity power factor.

OPERATION (Continued)

INITIAL OPERATION (Continued)

4. 50 Hertz Bandpass Filter Tests: Check out the 50 Hz Bandpass Filter board (LOC 1103). The 50 Hz Bandpass Filter is used to remove excessive power system noise before insertion into the Frequency Transducer. The only adjustment on this board is the "Q" of filter P1. Using an oscillator, insert a signal from LA2-B to common. The signal should be approximately 3.5 VAC at 50 Hz. The gain through the 50 Hz Bandpass Filter should be 1.2 $\pm$ 0.2. Determine the upper frequency at which the response is down by 3dB, which is 0.707 times the value at 50 Hz. Determine the lower 3dB frequency by the same procedure. The "Q" of the filter is defined as $Q = f_o/(f_2 - f_1)$, where f_o is the center frequency at 50 Hz, f_2 is the upper 3dB frequency and f_1 is the lower 3dB frequency. The "Q" has been factory adjusted to be 3.0. If the measurements indicate that the "Q" deviates by more than 10% from the desired value, adjustment should be made using P1. This is an adjustment that can be made through a hole in the faceplate. It should be noted that an adjustment in "Q" affects the gain at center frequency so that a new center frequency gain should be used in determining the frequency at which the gain is reduced by 3dB.

5. Frequency Transducer Tests: Check out the Frequency Transducer board (LOC 1101). Pull the 50 Hz Bandpass Filter (LOC 1103). Check that A1S shorting clip on the Frequency Transducer board is on terminals 2 and 3. Using an oscillator, insert a signal into IN test jack located on the faceplate. Adjust the signal for 50 Hz having an amplitude of approximately 3.5 VAC. The output of the frequency transducer should be 0 $\pm$ 0.1 VDC. Do not make any adjustments in this null unless the output voltage deviates from the limits given and the frequency of the oscillator is accurately known. Adjustment can be made later with the generator on line at rated voltage by using an Extender board. Change the frequency by $\pm$ 0.5 HZ ($\pm$ 1%) and note the output voltage. The gain should be 1 volt/percent so the total change in output should be + 1 - (-1) = 2. Do not adjust the gain unless it deviates by more than 10% from 1 volt/%. Change switch A1S to 1, 2. The gain should now be 0.1 volt/% $\pm$ 10%. Change switch A1S back to 2, 3. The frequency transducer output should be positive for an increase in frequency and negative for a decrease in frequency (assuming the output is at zero volts at 50 Hz). A "Jog" pushbutton is provided which inserts a step into the PSS for test purposes. With 50 Hz on the input to the frequency transducer, press the "Jog" pushbutton. The output voltage should decrease by approximately 5 mv. Release the pushbutton and the voltage should return to the original value.

6. Frequency Response Tests: If it becomes necessary to check the frequency response of the PSS, set Signal Conditioning board (LOC 1102) for lead time constants of 0.26 seconds and lag time constants of 0.03 seconds.

OPERATION (Continued)

INITIAL OPERATION (Continued)

6. Frequency Response Tests: (continued)

This can be accomplished by the following adjustments:

TCT2V	=	0 turns
TCT2F	=	0.9 turns
TCT3	=	1.5 turns
TCT4	=	0.9 turns
TCT5	=	1.5 turns

The twin-tee notch filter should already be set for 50 Hz. On the Washout and Output board (LOC 1108) set the washout time constant for 2 seconds and the gain for 20. This can be accomplished by placing the TCT1 dial on 6.2 turns and the gain control on 3.8. Jumpers on C1S, C2S and C5S should be on 2-3. Those on C3S and C4S should be on 1-2.

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THE PSS MUST BY OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

Pull the Frequency Transducer board (LOC1101). Insert the Extender board in place of the Frequency Transducer board. Connect an oscillator to terminal 15 or 16 and terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS from 0.1 Hz to 100 Hz measuring oscillator output and output test connector on the Washout and Output board. The oscillator output level should be held at 100 mv zero-to-peak, or 200 mv peak-to-peak, to avoid saturation of the amplifiers. Compare gain and phase to values shown in Figure 1.

7. Ceiling Sensing Tests: Check out the Ceiling Sensing board (LOC 1110). Pull the Ceiling Sensing board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and the negative limit (E3P) for 500 (5.0 turns). Set the time delay on the on the Ceiling Sensor board for minimum (0). Insert a variable dc voltage into IN1 of the Washout and Output board. Place the

OPERATION (Continued)

INITIAL OPERATION (Continued)

7. Ceiling Sensing Tests: (continued)

regulator in the manual mode of operation.

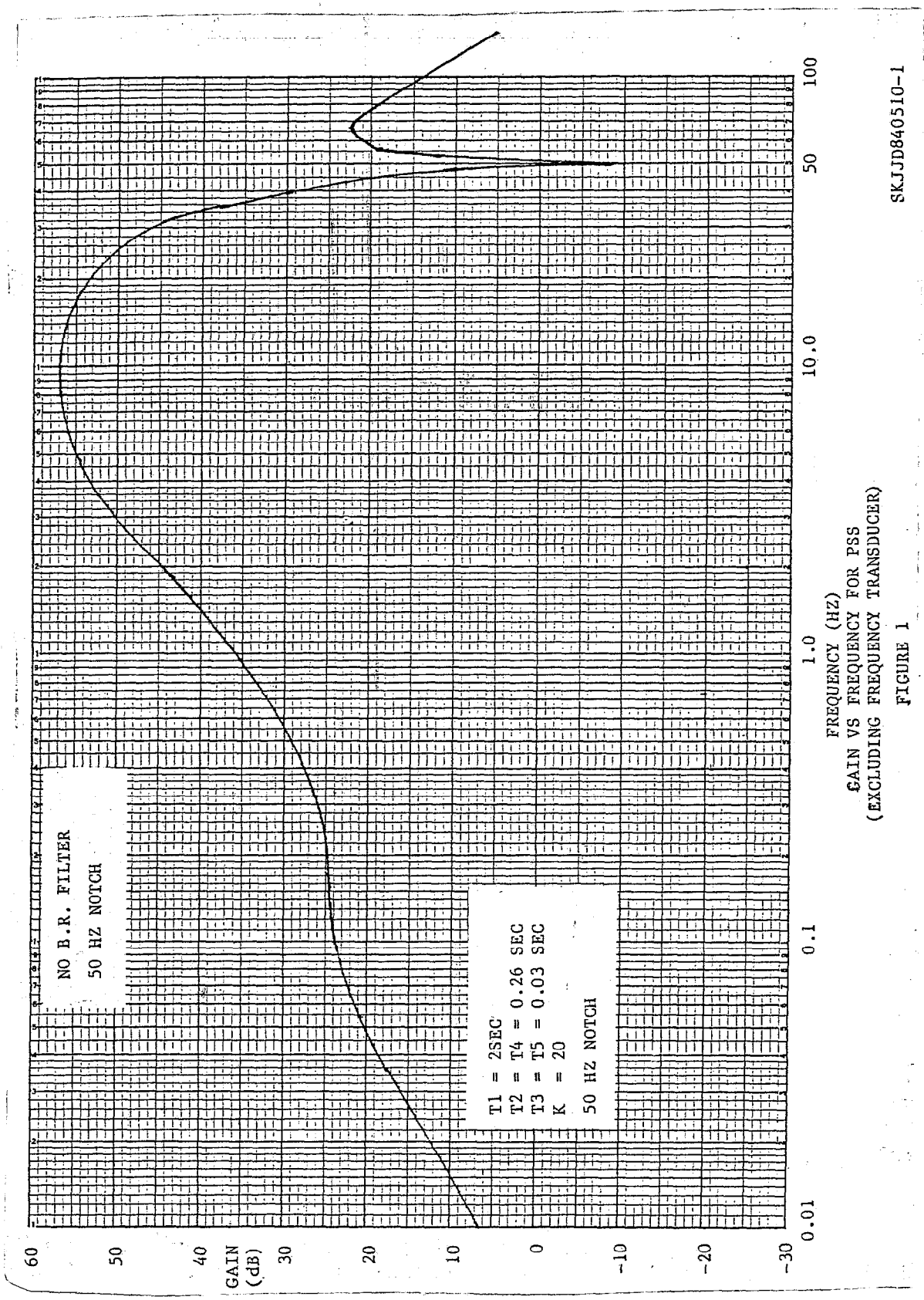
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* *
* CAUTION *
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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS COULD RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" pushbutton or LED being lighted. The voltage at Out 1 on the Washout and Output board should be 5 volts ± 0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working.

Adjust the time delay (E1P) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Remove the test voltage and reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero.

8. Washout and Output Tests: On the Washout and Output board (LOC 1108) set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable dc voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the dc voltage slowly and note at what voltage the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts ± 0.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude (within ± 0.5 volts). Remove the voltage source and Extender board. Reinsert the Frequency Transducer board. Ensure that the PSS is off. Place the regulator in the automatic mode of operation.



GAIN VS FREQUENCY FOR PSS
(EXCLUDING FREQUENCY TRANSDUCER)

FIGURE 1

OPERATION (Continued)

INITIAL OPERATION (Continued)

9. Maximum Gain Noise Tests: The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T3 set at 1.5 and 0.02 seconds, respectively, the gain will be 116 dB (631,000). Although this gain will probably never be used in an actual system, a good check of PSS noise level can be obtained using this gain. Ensure that the PSS is off. On the Signal Conditioning board (LOC 1102), set TCT2F and TCT4 for 1000 (10 turns) and TCT3 and TCT5 for zero. Pull the Washout and Output board (LOC 1108). Put the shorting clip on C5S on terminals 2 and 3 (GRD). Put the shorting clip on C3S on terminals 2 and 3. Insert Washout and Output board. Turn gain adjustment to 1000 (10 turns). Remove the Frequency Transducer board (LOC 1101). Insert Extender board in place of the Frequency Transducer. Short from pin 15 to pin 3 of the Extender board. Using an oscilloscope with a very slow sweep (1 sec/cm), determine the peak-to-peak output voltage (measured at OUT 1 of the Washout and Output board). This voltage should be observed for some time, perhaps 10 seconds to 1 minute. This voltage should be less than 3 volts peak-to-peak. Remove Extender board. Insert Frequency Transducer board in LOC 1101. Remove Washout and Output board. Place C5S shorting clip on terminals 1 and 2 (not GRD). Place C3S shorting clip on terminals 1 and 2. Return gain adjustment dial to zero (unity gain). Insert Washout and Output board. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

10. Optional Circuit Tests: Check out the following optional circuits if supplied with this equipment.

    A. Voltage Sensing Panel Tests - Check out the Voltage Sensing panel located below the printed circuit board rack. Remove the Voltage Deviation Sensing board and the Under-Overvoltage Detection board. Apply 120 VAC to L25, L26 and L27 (ØA, ØB and ØC, respectively). Close the isolating switches mounted on the "L" terminal board at L25, L26 and L27. The voltage from LB5-B to LB5-A should be 38 VDC ± 10%. Replace the Voltage Deviation Sensing and Under-Overvoltage Detection boards.

    B. Under-Overvoltage Detection Tests - Check out the Under-Overvoltage Detection board (LOC 1109). Put the Under-Overvoltage Detection board on an Extender board. Measure the voltage at D5TP. If the voltage is not + 5 VDC ± 0.05 VDC, then adjust using D4P. Measure the voltage at D6TP. If the voltage is not - 5 VDC ± 0.05 VDC, then adjust using D6P. Measure the voltage at D2TP. If the voltage is not - 5 VDC ± 0.05 VDC, then adjust using D2P. At an input voltage corresponding to rated, measure and record the voltage at D1TP. Adjust D1P (input voltage) for zero volts at OUT 1. Place C1S

OPERATION (Continued)

INITIAL OPERATION (Continued)

10. B. Under-Overvoltage Detection Tests - (continued)

shorting clip on terminals 1 and 2 of the Washout and Output board. The voltage at OUT 2 should be > 2.0 VDC. Adjust the positive and negative limit dials to zero (0 turns). Change the generator voltage by approximately 1% (or change the voltage at pins 15 and 16 by 1% if a dummy voltage source is used). Turn the appropriate limit dial. If voltage was raised, use the positive limit and if lowered, use the negative limit. Determine the setting at which OUT 2 changes to approximately 0.5 volts. The dial setting should be 100 ± 5. When the terminal voltage is changed back to nominal, the voltage at OUT 2 should return to > 2.0 VDC.

With the C5S shorting switch on terminals 1 and 2 (on the Washout and Output board), insert a + 1 VDC signal into IN1 of the Washout and Output board. If noise is present in the output, it may be necessary to pull the Signal Conditioning board. With the gain dial set for zero, the voltage at OUT 1 on the Washout and Output board should be - 1 VDC. Change terminal voltage by an amount sufficient to cause the voltage at OUT 2 of the Under-Overvoltage Detection board to change to approximately 0.5 VDC. The voltage at OUT 1 on the Washout and Output board should be reduced to zero volts (± 0.1 VDC). Change terminal voltage back to nominal and the voltage at OUT 1 on the Washout and Output board should return to - 1 VDC after approximately 0.5 seconds. Remove the signal from IN1 of the Signal Conditioning board. The positive and negative limit dials should be set to the values specified by the customer. C5S should be left on terminals 2 and 3 and C1S should be left on terminals 1 and 2.

C. Voltage Deviation Sensing Tests - Check out the Voltage Deviation Sensing board (LOC 1107). Put the Voltage Deviation Sensing board on the Extender board. Adjust the gain dial for 1000. Place B1S on the Signal Conditioning board on terminals 2 and 3. Measure the voltage at H1TP. If this voltage is not - 5 VDC ± 0.05 VDC, then adjust using H2P. Place H1S shorting switch on terminals 2 and 3 (GRD). With rated generator terminal voltage (or with a source of dc voltage connected from pin 11 to pin 13 and adjusted for approximately 30 VDC) adjust H1P for zero volts at OUT. Change terminal voltage by 1% (or the dc source by 1%). The voltage at OUT should be 0.26 ± 0.025 VDC. If an external source is used to simulate a terminal voltage signal, the signal should be inserted between pins 11 and 12 and between pins 13 and 14. Open the PT isolation switches at L25, L26 and L27 if an external source is used. Note that the Voltage Deviation Sensing board causes an inversion of the signal.

OPERATION (Continued)

INITIAL OPERATION (Continued)

10. C. Voltage Deviation Sensing Tests - (continued)

Return the gain dial to zero, remove the external voltage source (if used) and close the PT isolation switches. Remove the Extender board and re-insert the Voltage Deviation Sensing board. B1S should be left on terminals 1 and 2 and H1S should be left on terminals 2 and 3.

11. Initial Settings:

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different systems, adjustments are made using calibrated dials. On the Washout and Output board, set the gain dial to zero and C3S on pins 1 and 2. Ensure that the Relay board has the correct rheostat and switch settings. Check the Frequency Transducer board to ensure that A1S is on the proper terminals for the desired gain (0.1V/% or 1.0V/%). If the "internal frequency" option is to be used, check that the shorting jumper across LA2-B and LA2-D is removed. If the optional Under-Overvoltage Detection board is being used, ensure that C1S has the shorting link on terminals 1 and 2. If the optional Voltage Deviation Sensing board is being used, ensure that B1S on the Signal Conditioning board has the shorting link on terminals 1 and 2. Close the PT isolation switch and the CT isolation switch (if the "internal frequency" option is being used.)

Settings for a particular installation are best obtained by analytical studies and/or on line tests. It is the responsibility of the customer to perform the analytical studies and on line tests necessary to determine proper settings. General Electric Company can provide assistance in performing analytical studies and system tests. Contact your local sales office. A paper, "Concepts of Synchronous Machine Stability as Affected by Excitation Control", by F. P. deMello and C. Concordia, may be used as a reference (68-TP-129-PWR). Other publications which may be used as references are "Test Procedure for Power System Stabilizers", published by the Western Systems Coordinating Council and an IEEE publication, "Guide for the Identification, Testing and Evaluation of the Dynamic Performance of Excitation Control Systems", paragraph 6.03 (P421, 1977).

12. On-Line Tests:

Power System Stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems; local machine system oscillations in the 1 - 2 Hz range and inter-area oscillations $\leq$ 0.5 Hz. The fundamental function of the signal conditioning network

OPERATION (Continued)

INITIAL OPERATION (Continued)

12. On-Line Tests (continued):

of the PSS is to compensate for the phase lags of the system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions (usually at or near full load). The desired stabilizing signal has as its reference the base system frequency which is slowly changing all the time. The washout function serves to minimize the effect of overall system speed changes on machine terminal voltage.

The PSS has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The output voltage limits are independently adjustable but should be set identically in order to prevent non-symmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable output limit stage, the output voltage limits should be set at a higher voltage than the ceiling sensing limits.

In order to facilitate on line frequency response testing of the excitation system, IN1 is provided on the Washout and Output board. With the shorting clip on C5S on terminals 1 and 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal operation, the shorting clip on C5S should be on terminals 2 and 3 (GRD), and the shorting clip on C1S should be on terminals 2 and 3.

Figure 2 is provided so that dial settings can be chosen to realize the desired time constants and limit settings.

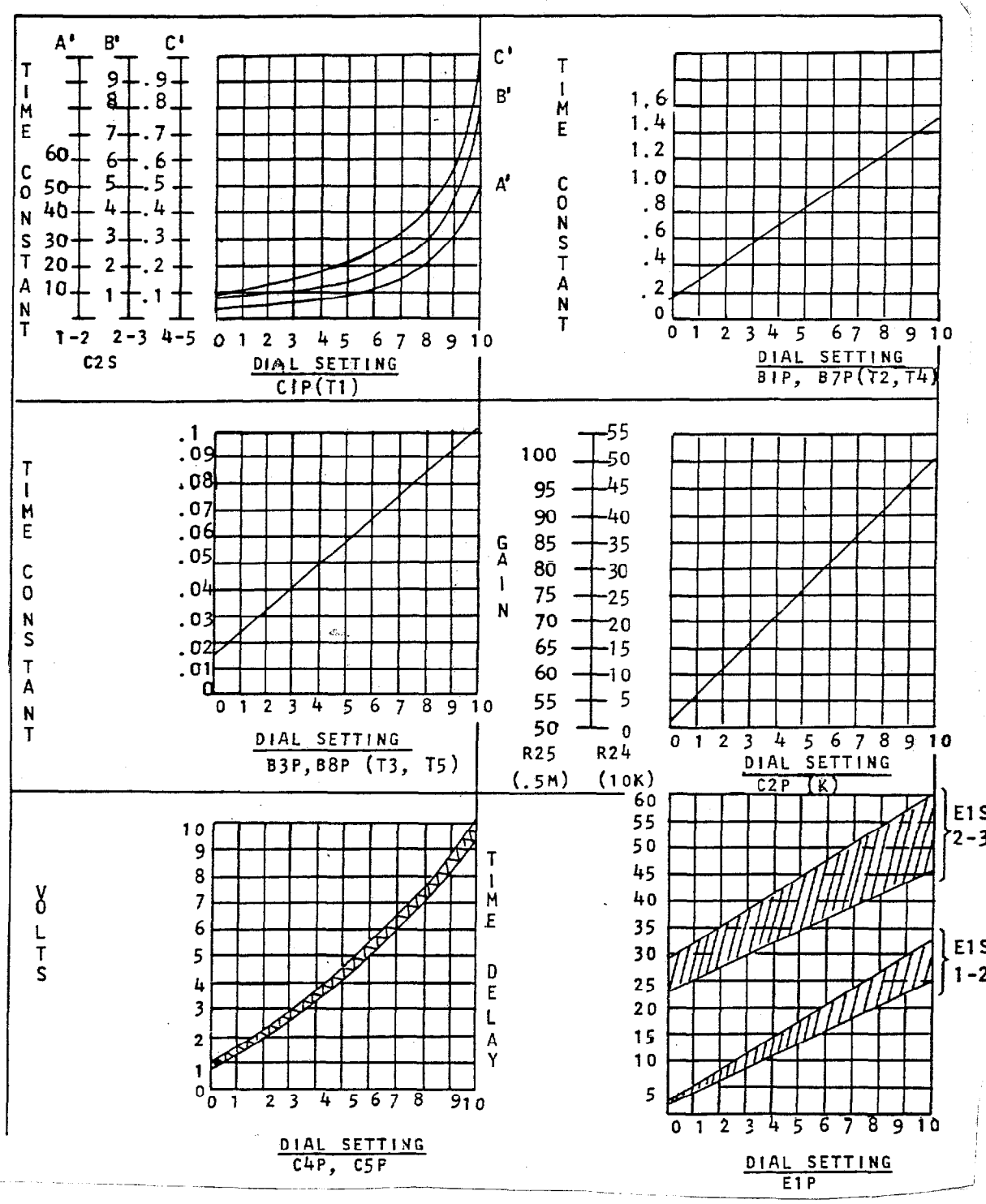


Figure - 2 - Parameter Value vs Dial Setting

PRINCIPLES OF OPERATION

GENERAL

The Power System Stabilizer is used to provide an additional input to the voltage regulator which, under certain system conditions, can improve power system dynamic stability. A signal is obtained from terminal voltage (or, optionally, terminal voltage added vectorially to line current). This signal is frequency modulated, with the carrier frequency being 50 Hz and the modulation frequency being equal to the frequency changes of terminal voltage. This frequency modulated signal is passed through a bandpass filter and frequency transducer to produce an output signal proportional to the magnitude of the deviation in frequency with a frequency equal to the deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensation to provide phase compensation for the system, a twin-tee notch filter tuned to 50 Hz, a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service (and also removal from service). Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. Protective circuits are provided to detect a failure which causes the output to go to ceiling (Ceiling Sensing board) and optionally a failure or system condition which causes terminal voltage to deviate by more than a preset value from nominal (Under-Overvoltage Detection board). Ten turn rheostats with calibrated dials are furnished for all customer adjustments. Figure 6 includes an overall block diagram of the PSS. Individual board and panel elementary diagrams are furnished with the equipment. It will be necessary to have these for much of the following explanations.

POWER SUPPLY SYSTEM

The panel mounted PSS includes both the ± 15 VDC and 24 VDC power supplies. The 115 VAC, 50 Hz power should be connected to L21 and L22. S1T and S2T transform the input voltage (and provide isolation) and then the voltage is rectified by a single phase full-wave bridge, filtered to remove the AC component, limited by Zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCRs R1CR and R2CR, along with associated sensing and firing circuits form a "crow-bar" circuit which will protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators (G1A, G2A) are used to control the output voltage. G2P and G4P mounted on he printed circuit board provide adjustment of the output voltage. G1R, G1P, G2R and G3R provide a current limit feature on the + 15 volt regulator (a similar circuit is found on the - 15 volt regulator). The current limit is of the foldback type, i.e.,

PRINCIPLES OF OPERATION (Continued)

POWER SUPPLY SYSTEM (Continued)

the maximum output current is reduced as the load on the regulator is increased above the full-load value. This ensures that critical components in the regulator (series pass transistors) will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 Amp and the negative limit at 750 ma (these are not the short circuit current values). Integrated circuits J1A and J2A provide voltage sensing for the "crow-bar" circuit. J1P and J2P are factory adjusted to cause the "crow-bar" circuit to operate at a supply voltage of 17 volts. S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. (Refer to the diagram for the correct tap for a given input voltage.) In order to reduce the noise level to as low a value as possible, the $\pm$ 15 volt power supply is very tightly regulated and heavily filtered. The regulation is 0.1% and the ripple on the power supply should be $\leq$ 5 mv peak-to-peak. The 24 volt supply does not require tight regulation since it is used to drive relays and energize lights. S3T is used to isolate the 24 volt supply from the $\pm$ 15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by a Zener diode (S1Z). The 24 volt power supply is ungrounded while the $\pm$ 15 volt supplies are grounded through a parallel resistor - capacitor combination. The high impedance ground on the $\pm$ 15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

INTERNAL FREQUENCY PANEL

The Internal Frequency panel contains potential transformers which accept a three-phase voltage proportional to generator terminal voltage (115 VAC) and generate a single phase voltage ($\approx$ 3.5 VAC) in phase with phase B line - to - neutral voltage. Taps are provided on the secondary of L1T and L2T to provide 2 VAC on each secondary with an input voltage which varies from 110 to 120 VAC. L1T and L2T are connected open delta with the polarity of L1T reversed on the secondary. This gives the proper phase shift so that when L1T voltage is added to L2T voltage, a voltage in phase with B phase line - to - neutral voltage is developed.

A reactor is provided which develops a voltage proportional to line B current (5 Amps at 1 per unit load) and phase shifted from line B current. If it is desired to use the so called "internal frequency" option, a shorting jumper is removed from the secondary of L1X and the CT shorting switch is closed. This gives a sensing voltage equal to $V_t + jIX$, the frequency of which, under most conditions, approximates the speed of the rotor. The reactance should be equal to X_Q of the generator. Figure 3 shows the phasor diagrams for generating a voltage having the so-called "internal frequency".

PRINCIPLES OF OPERATION (Continued)

INTERNAL FREQUENCY PANEL (Continued)

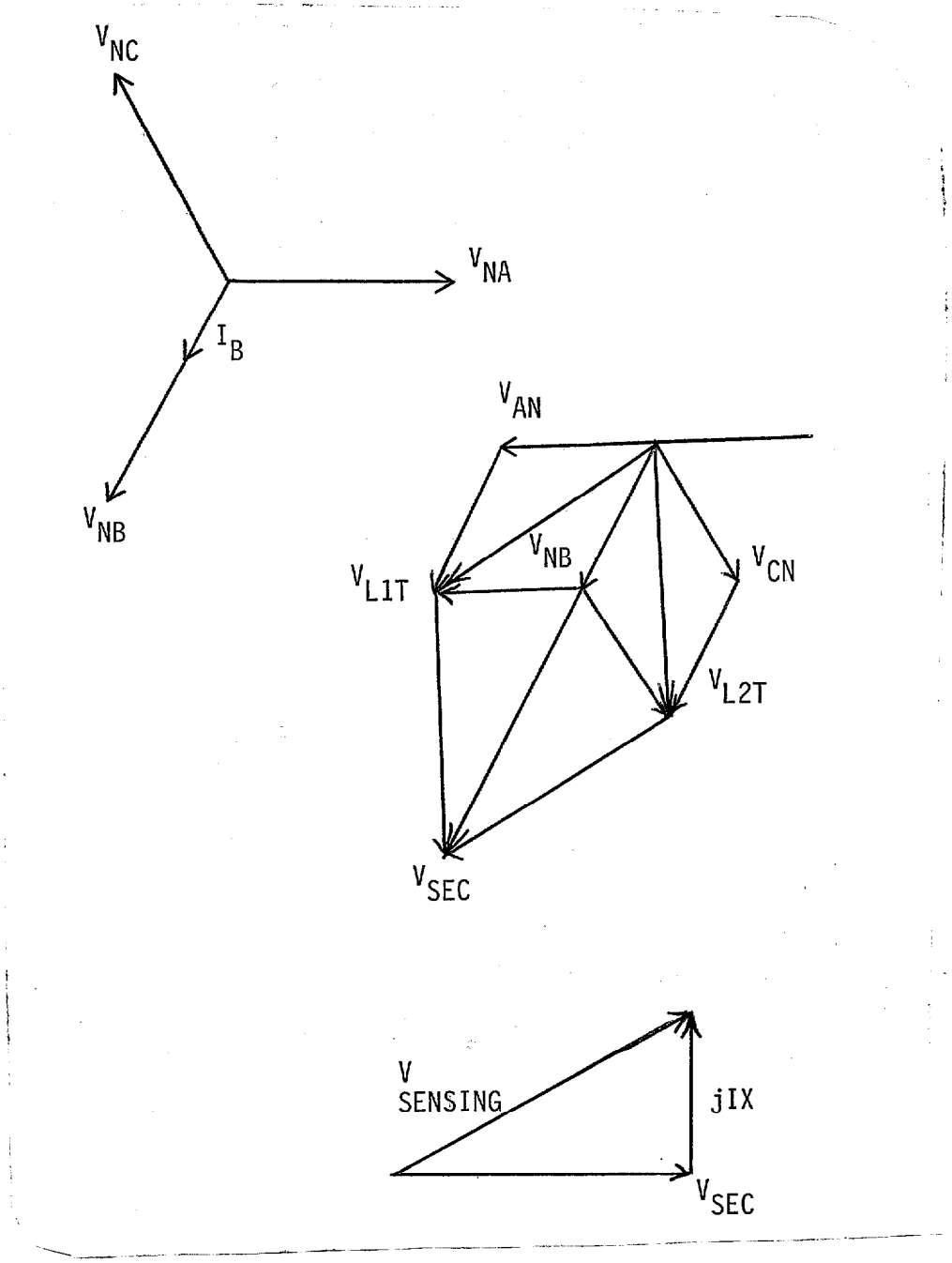


Figure - 3 - Phasor Diagrams for Generating a Voltage Having the so-called "internal frequency"

PRINCIPLES OF OPERATION (Continued)

FREQUENCY TRANSDUCER BOARD (Continued)

to sample the peak of the integrator output (proportional to the time of a half-cycle) and hold this value until the next negative half-cycle discharges the integrating capacitors and resets the integrator. The output of the sample and hold device is fed to A3A which also has a bias signal of the opposite polarity to give zero volts at rated frequency. A jog pushbutton is provided which causes the output of the Frequency Transducer to change (-) 0.005 volts when pressed. A4A develops adequate gain for either 1 volt/% change in frequency or 0.1 volt/% change in frequency selected by jumper A1S mounted on the printed circuit board.

SIGNAL CONDITIONING BOARD

The Signal Conditioning board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the Frequency Transducer, an extra input with separate lead time constant adjustment is provided for a voltage input (IN2). B1P provides adjustment of the first stage lead time constant which is proportional to B1C, B1R and the setting of B1P. B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The dc gain of the stage is B9R/B1R which is 1.0. The B3R, B3C and the equivalent resistance of B3P cause extraneous lead and lag time constants, which have been minimized, but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage limit circuit formed by B1Z, B2Z, B5D, B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D to increase the output current capability of B2A. A twin-tee notch filter is placed between the first and second lead-lag stages to remove 50 Hz. The twin-tee is factory adjusted to produce greater than 50 dB attenuation at the notch frequency. The notch frequency is given by:

$$f = \frac{1}{2\pi(B11R + B4P)(B16C)}$$

PRINCIPLES OF OPERATION (Continued)

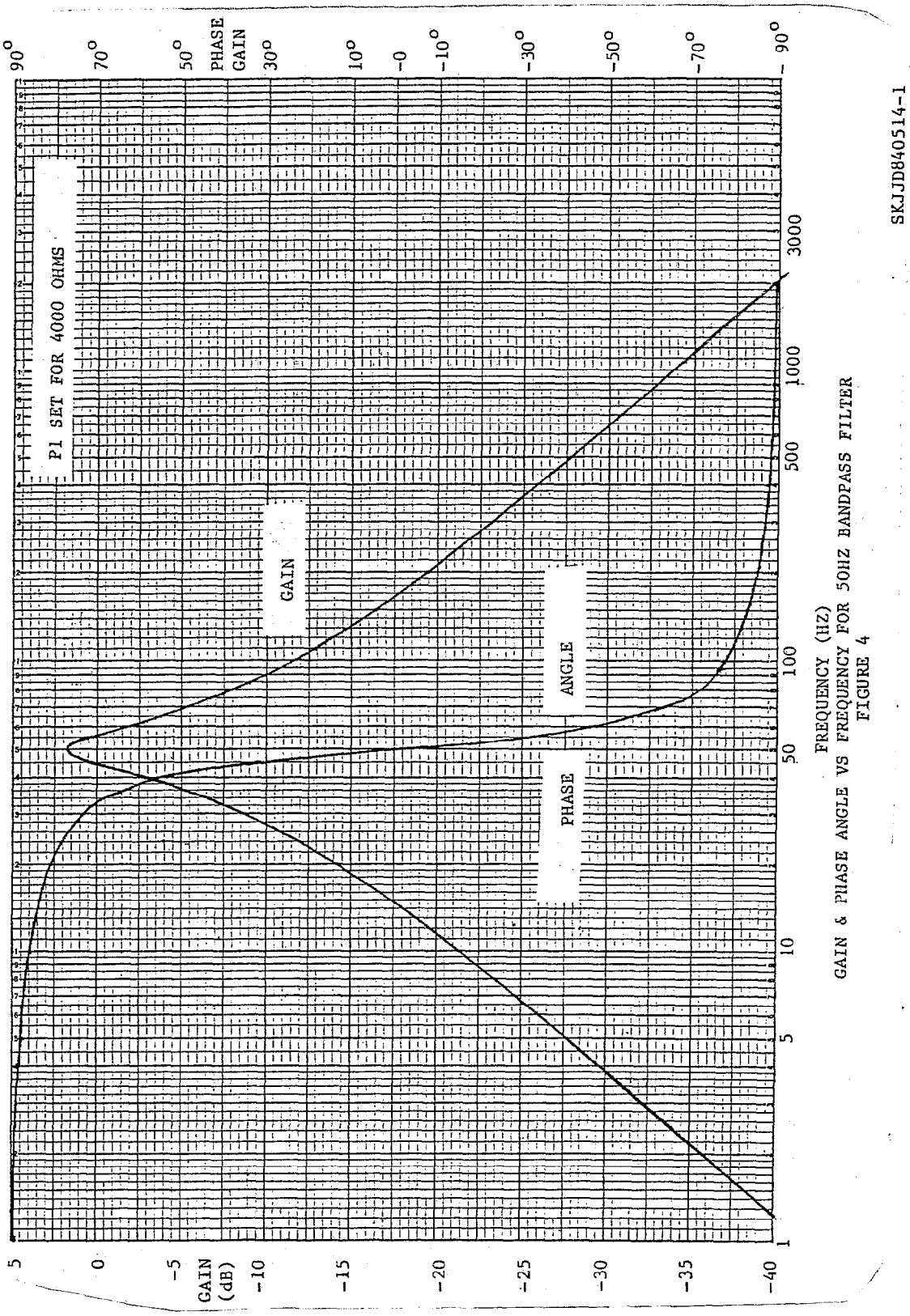
50 HERTZ BANDPASS FILTER BOARD

Tests on a PSS using a frequency transducer to demodulate a terminal voltage signal indicate that excessive power system noise can render the PSS inoperative. By providing an input filter to remove the higher frequency noise components, acceptable noise levels have been obtained. The bandpass filter is positioned to filter the combined voltage and current signal (if a current signal is used). A single operational amplifier (with associated resistors and capacitors) is used to construct the bandpass filter. This realization gives no inversion of the signal and offers a relatively high degree of insensitivity to component drift. Center frequency is determined by 1% capacitors and 1% resistors which give a $\pm$ 2% tolerance of center frequency in the worst case. The "Q" of the filter which is defined as $Q = f_o/(f_2 - f_1)$, where f_o is the center frequency, f_2 is the upper frequency at which the amplitude is down 3dB from the amplitude at center frequency and f_1 is the lower frequency at which the amplitude is down 3dB from the amplitude at center frequency, is set for 3.0. This has provided adequate noise reduction while causing only 3 degrees of phase lag in a 0.5 Hz signal modulating a 50 Hz carrier and about 10 degrees phase lag when the modulating frequency is 1.5 Hz. Fine adjustment of the "Q" is provided by P1 which is a screwdriver adjustment through a hole in the faceplate. The gain of the filter is proportional to "Q" and so 1R and 2R form a voltage divider circuit to give an overall gain at center frequency of approximately 1.2. The phase shift through the filter at center frequency is zero. This provides an accurate method for determining center frequency. Figure 4 shows the frequency response of the filter.

FREQUENCY TRANSDUCER BOARD

The Frequency Transducer board (LOC 1101) produces a signal proportional to the frequency of an input voltage. A single phase voltage proportional to phase B voltage (or synthesized generator internal voltage) is fed to a crossover detector (A1A and associated circuitry) which produces square waves having constant amplitude and a period equal to the period of the incoming signal. Hysteresis is provided by A2Q, A3Q, A3D, A4D, A5D, A6D and A1Z to prevent chattering of the crossover detector (42 mv of hysteresis). A2Z is used to provide a precise amplitude signal to integrator A2A. A5Q is a constant current source to further stabilize the voltage on A2Z. A6Q is used to force the voltage across A2Z to zero during the positive half-cycle of input voltage.

On the negative half-cycle of input voltage, A2Z develops a constant voltage and this is integrated by A2A. At the beginning of the positive half-cycle, A2Z will have zero volts and a one-shot will trigger a sample and hold device



GAIN & PHASE ANGLE VS FREQUENCY FOR 50HZ BANDPASS FILTER
FIGURE 4

PRINCIPLES OF OPERATION (Continued)

WASHOUT AND OUTPUT BOARD

The Washout and Output board contains a washout function, a gain function and a power amplification function. Operational amplifier C1A is a chopper stabilized device for low offset. C1A is basically a derivative circuit formed by C1C and the feedback resistor (C2R, C3R or C4R). C2C is added to make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. C9D, C10D, Z1C, Z2C, C31R and C32R form a clamp to limit the output voltage of the washout stage. An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1 and 2 allows IN1 to be used for driving the regulator. C2A is used as an adjustable gain stage with the gain being determined by C3S and C2P. C3S is used to select a range of gain from 1 to 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal for Ceiling Sensing board through pins 13 and 14. C3Q and C4Q are junction FETs used as a static switch. IN3 can be connected to an optional Under-Overvoltage Detection board which serves to disconnect the PSS whenever terminal voltage deviates by an adjustable amount. In normal operation, C1S should have the shorting clip between terminals 2 and 3 (OUT). C3Q will be conducting and C4Q will be non-conducting. C3A is connected to a push-pull transistor stage (C7Q, C8Q, C9Q and C10Q) in order to increase the output current capability to $\pm$ 100 ma. C3A and the push-pull stage are connected as a unity gain voltage follower.

RELAY BOARD

The Relay board is used to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter M1 is mounted on the faceplate and can be used to ensure that the PSS output is zero before insertion into the voltage regulator. OFF and ON pushbuttons are mounted on the faceplate and can be used to insert or remove the PSS from service. Lights mounted on the board or in the pushbuttons indicate the status of the PSS. A line breaker contact (52G) should be used as a permissive contact so that the PSS is inoperative off line. The pushbuttons have momentary contacts and so F93 is sealed in by an FAR and F93 contact. The F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the Ceiling Sensing board.

PRINCIPLES OF OPERATION (Continued)

RELAY BOARD (Continued)

F1R, F2R and F1P are used to properly scale the PSS signal for insertion into the regulator. F2P is used to dummy load the regulator so that a bump in excitation will not occur whenever the PSS is placed in service. (Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S and F4S.)

CEILING SENSING BOARD

The Ceiling Sensing board is used to detect a failure in the PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the Ceiling Sensing board.

E1A and E2A operate as comparators with the E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit (E2P) and more positive than the negative limit (E3P) then the voltage at E5TP is at a minimum and the voltage at E6TP is at a maximum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its output is in the low state. The output of this nand gate is passed through two inverters (no change in state) and is also used to trigger a one-shot (E3L). With a ceiling condition not detected, the one-shot trigger input will stay in the low state and the output (the complemented output is used in this case) stays in the high state. The inputs to the first nand gate if E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected. E8C is provided to ensure that the cross-coupled gates will go to the proper state upon energization. With no ceiling condition detected, the inverter output of E2L will be in the low state turning on transistors E2Q and E3Q and energizing relay FAR on the Relay board.

If a ceiling condition exists (whether positive or negative), the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot (E3L) and changes the input on the first nand gate of E2L to a high state. Nothing happens, however, because the output of the one-shot (E3L) is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R (depending on the setting of E1S). After the timing interval, the output of the one-shot (E3L) goes to the high state and the output of the

PRINCIPLES OF OPERATION (Continued)

CEILING SENSING BOARD (Continued)

first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state which turns off E2Q and E3Q and drops out relay FAR on the Relay board, giving an alarm. To clear the alarm and place the PSS back in service, it is necessary to press the reset pushbutton (E1PB). This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared.

OPTIONAL EQUIPMENT

A. Under-Overvoltage Detection Board (Optional)

The Under-Overvoltage Detection board (LOC 1109) is used to remove the PSS from service whenever generator terminal voltage deviates by a preset amount from nominal. D2P is used to establish a reference for the input voltage signal and ensure proper scaling. D1P is used to null out the reference voltage at rated terminal voltage so that a change from rated is detected. A faceplate mounted test point (IN2) is provided for insertion of a test signal. D2A and D3A are used as comparators with D3P and D4P establishing a variable reference point. D4P and D6P provide proper scaling so that D3P and D4P dial setting (0 to 10) corresponds to percent voltage deviation from nominal. The output normally stays at $>$ 2.0 VDC with no undervoltage or overvoltage detected. When and undervoltage or overvoltage is detected, the output goes to $\approx$ 0.5 VDC. If terminal voltage changes back to a value less than the limits, the output changes back to $>$ 2 VDC after 1/2 second time delay. The output of the Under-Overvoltage Detection board is used to control a static switch on the Washout and Output board. If the Under-Overvoltage Detection board is to be used, then the shorting link on C1S must be placed between terminals 1 and 2. Nominal input voltage is 30 VDC at rated terminal voltage.

B. Voltage Deviation Sensing Board (Optional)

The Voltage Deviation Sensing board (LOC 1107) is used to provide an input to the PSS proportional to the deviation of terminal voltage from rated. The input signal is obtained from the Voltage Sensing panel and is nominally 30 VDC at rated generator terminal voltage. H2P is used to provide a reference voltage and to allow proper scaling of the input signal. H1P is used to null out the reference voltage at rated terminal voltage. A faceplate mounted test point (IN2) allows insertion of a

PRINCIPLES OF OPERATION (Continued)

OPTIONAL EQUIPMENT (Continued)

B. Voltage Deviation Sensing Board (Optional)

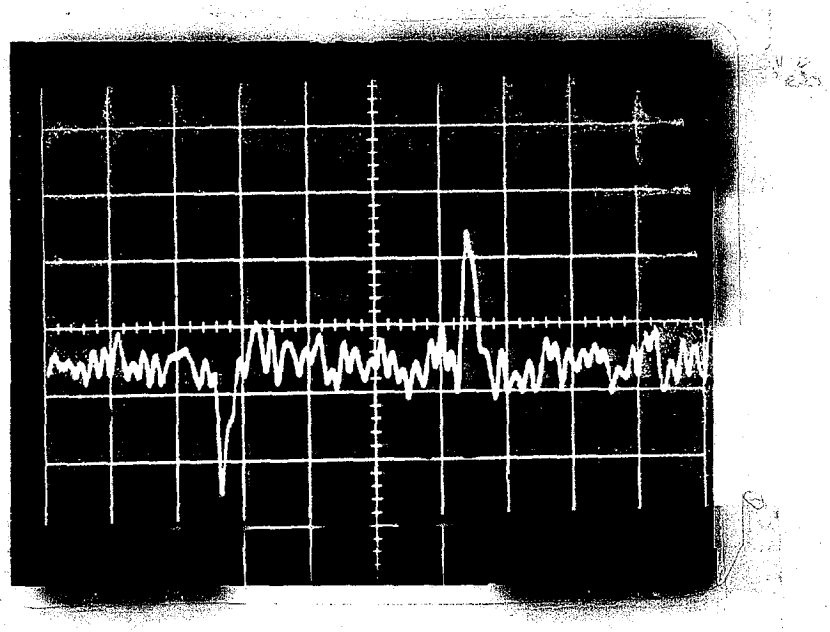
signal for test purposes. Gain adjustment is provided for a range of 0.2 to 5.2 volts/volt or 0.01 to 0.26 per unit/per unit. If the Voltage Deviation Sensing board is to be used, the shorting link on B1S on the Signal Conditioning board must be on terminals 2 and 3, (not GRD).

NOTES

MAINTENANCE

The PSS is a static device having no moving parts except for relays and instruments. Therefore, little maintenance should be required. If excessive vibration is present, all screw connections should be checked periodically. The Frequency Transducer null adjustment should be checked periodically and readjusted if necessary. A pushbutton is furnished on the Frequency Transducer which puts a small step into the Frequency Transducer output. The step is equivalent to approximately 0.005% change in frequency. This change is so small that system noise may mask the effect on some systems. If a response can be observed in the output of the Washout and Output board when pressing the pushbutton, record for future reference. A quick check on the operating status of the PSS can then be made by pressing the pushbutton and comparing the output with the recorded output. The following photograph (Figure 5) shows the output of the Washout and Output board when the pushbutton is pressed and then released.

($T_2 = T_4 = 0.5$ sec; $T_3 = T_5 = 0.05$ sec; $T_1 = 10$ sec and $K = 1.0$)



PSS JOG TEST

VERTICAL = 100 mv/division
HORIZONTAL = 200 msec/division

Figure - 5

TROUBLESHOOTING

1. Carefully check all wiring if problems are experienced at start-up.

2. If no response is obtained while making the frequency response test:

   a. Check that the power switch on the Power Supply board is on.

   b. Check fuses S1F and S2F on the Power Supply panel.

   c. Check that all boards are fully inserted.

   d. Check for proper jumper connections on the Washout and Output board.

   e. Check each board individually for a response.

3. If an incorrect signal is obtained in terminal voltage, check the switch and rheostat settings on the Relay board.

4. If the PSS will not stay in service:

   a. Press reset pushbutton on Ceiling Sensing board.

   b. Check that permissive contact is closed.

5. Frequency Transducer output stays at ceiling:

   a. Ensure that generator speed is at rated.

   b. Ensure that PT isolating switch is closed.

   c. Readjust null.

6. An instability is observed with gain dial at 0:

   a. Measure gain to ensure that it is unity.

   b. Ensure that an increase in frequency produces an increase in generator field voltage.

7. If a board is found to be defective, replace it and return the defective board for repair and return.

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location. Board components should not be replaced, order the entire board.

2. Complete nameplate data appearing on the assembly that included the component (including the serial number of the assembly).

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

NOTES

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932LA202

FUNCTIONAL ELEMENTARY

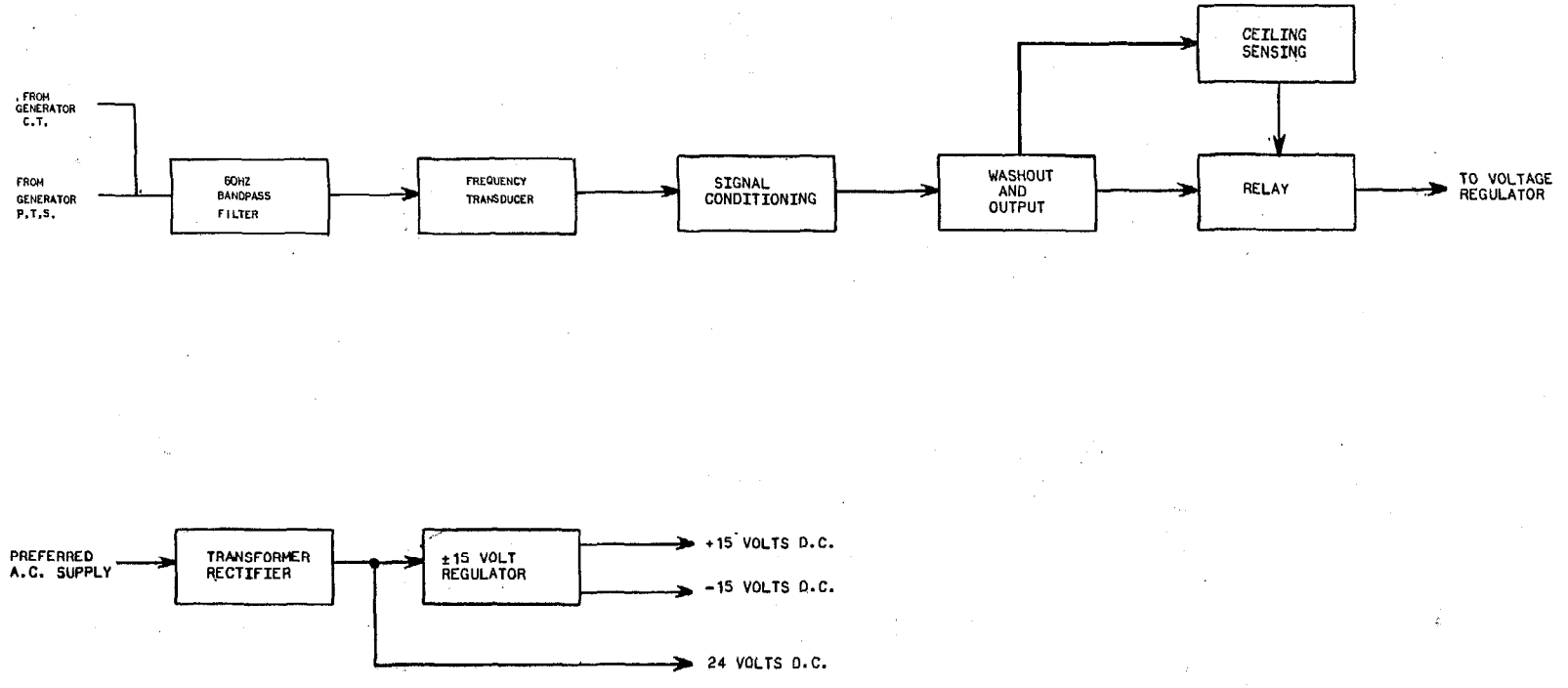


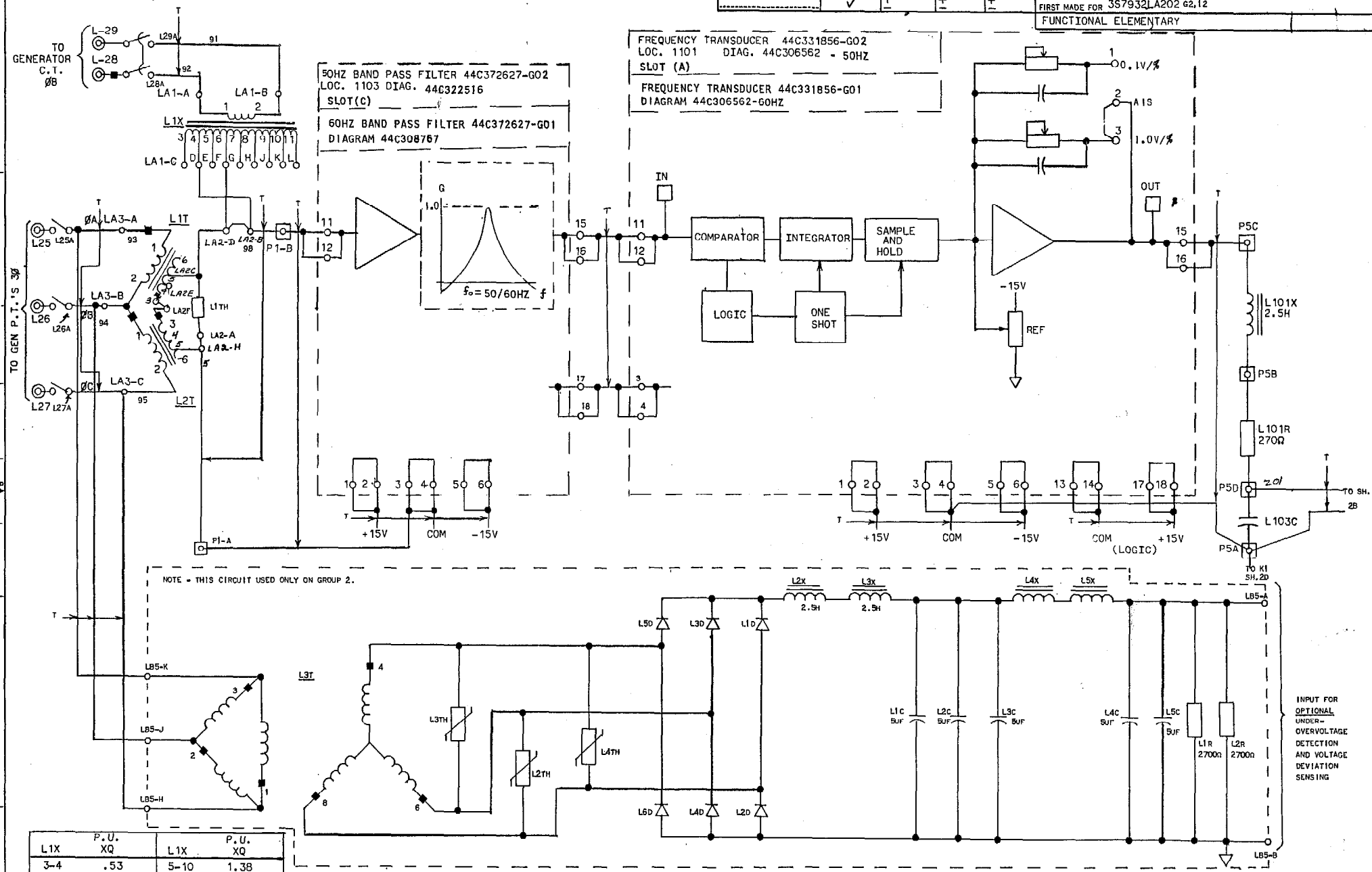
Figure 6, Sheet 1

REVISIONS			PRINTS TO
	1	NR 7-24-80 ADDED PRT. KEY	* 3EH1
			3QA1
			4QA5
			3QA2
			3QA3

MADE W. D. SLAUGHTER	APPROVED	DSD	DIV OR DEPT	3S7932LA202
ISSUED 4-6-79		SALEM, VA	LOCATION	CONT ON SHEET 2A SH NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202 G2,12
FUNCTIONAL ELEMENTARY



L1X	P.U. XQ	L1X	P.U. XQ
3-4	.53	5-10	1.38
4-5	.53	5-11	1.49
5-6	.53	4-7	1.6
6-7	.53	4-8	1.71
6-8	.64	4-9	1.81
6-9	.74	4-10	1.92
6-10	.85	4-11	2.03
3-5	1.06	3-7	2.13
4-6	1.06	3-8	2.23
5-7	1.06	3-9	2.34
5-8	1.17	3-10	2.44
5-9	1.27	3-11	2.55
5-10	1.38	6-11	0.96

Figure 6, Sheet 2

REVISIONS			PRINTS TO
7 ECC 2-8-82 DCS 2-9-82	1	NR 5-17-80 CHG. 60 HZ TO 50 HZ	4QA5
	2	NR 5-29-80 DWG 861B81 REV C	3QA1
	3	NR 6-30-80 CHG. REF.	3EH1
	4	NR. 7-24-80 ADDED PAT. KEY	
	5	NR 8-22-80	
	6	NEW 11-11-80	

MADE BY W.D.SLAUGHTER ISSUED 4-6-79
APPROVALS
DRIVE SYSTEMS PROD. DEPT. SALEM, VA
3S7932LA202 CONT ON SHEET 2B SH NO. 2A

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± 1 | DECIMALS ± | ANGLES ±

TITLE
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202 82.12
FUNCTIONAL ELEMENTARY

SIGNAL CONDITIONING - 44C331857-G03 - 50HZ
LOC. 1102 DIAG. 44C306563
SLOT (B)
SIGNAL CONDITIONING 44C331857-G01-60HZ

FROM SH. 2A
IN1
60HZ/ 50 HZ NOTCH
OUT
LEAD (T2-F)
LAG (T3)
LEAD (T4)
LAG (T5)
FOR USE WITH OPTIONAL VOLTAGE DEVIATION
B1S
LEAD (T2-V)
+15V COM -15V

CEILING SENSING - 44C331861-G01
LOC. 1110 DIAG. 44C306567
SLOT (F)
-15V
POS. LIMIT
NEG. LIMIT
COMPARATOR
COMPARATOR
LOGIC
RESET
4.7V
25 → 50 SEC
2 → 25 SEC
E1S
TIME DELAY
RELAY DRIVER
OUT
403 TO SH. 2C
+15V COM -15V

WASHOUT AND OUTPUT - 44C331859-G01
LOC. 1108 DIAG. 44C306565
SLOT (D)
IN1
IN2
C5S
C3S
1 → 50
50 → 100
C2S
8-50
1-8
.1-1
TC T1
POS. LIMIT
NEG LIMIT
-15V
+15V
STATIC SWITCH
LOGIC
OUT1
N1
L102X
1.5MH
N3
.0.1UF
L104C
N2
402 TO SH. 2C
TO K1 SH. 2D
C1S
(OPTIONAL UNDER-OVERVOLTAGE DETECTION INPUT)
+15V COM -15V

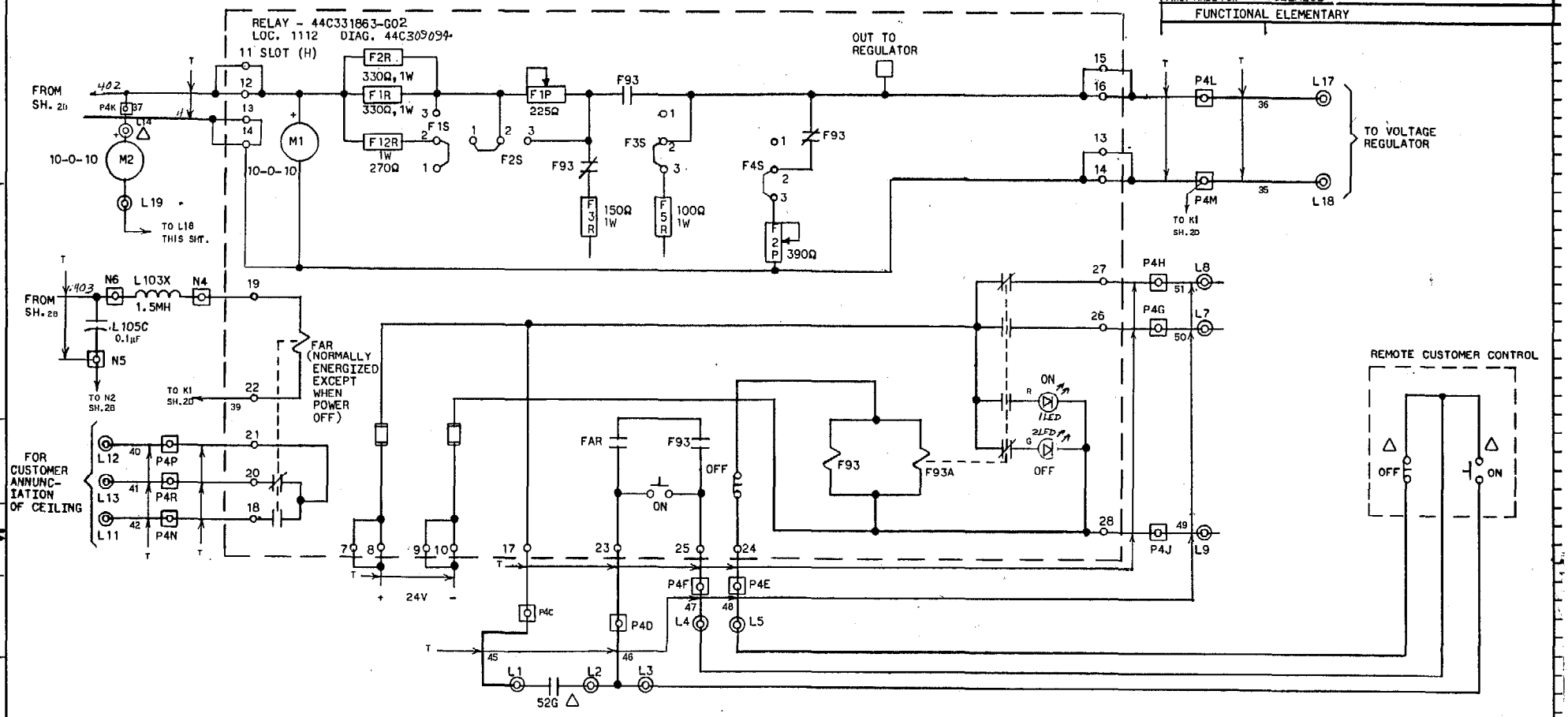
REVISIONS

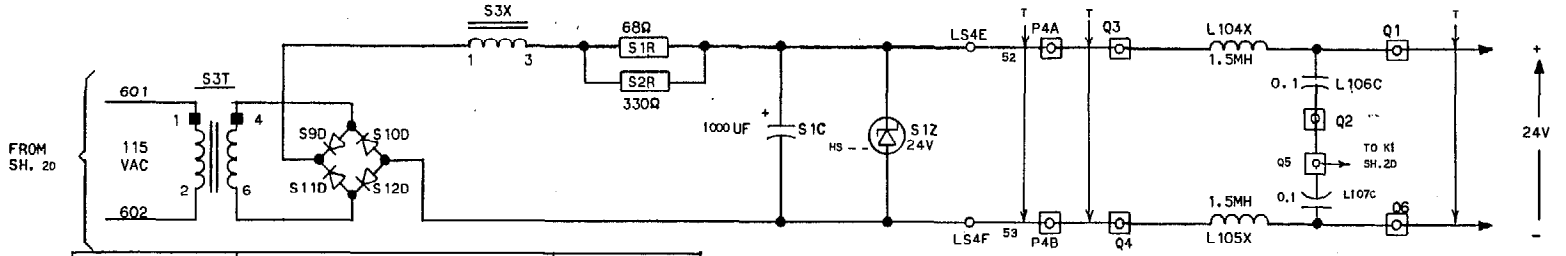
		PRINTS TO	
1	NR 6-29-80 84 11661 ARC	3QA2	3EH1
2	NR 6-30-80 Wi. Rect.	3QA3	4QA5
3	NR 7-24-80 Added Pot. Key		3QA1
4	ECL 8-8-84 SMD 2-9-84		

MADE BY WD SLAUGHTER
ISSUED 4-6-79
APPROVED
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
SALEM, VA. LOCATION
3S7932LA202 CONT ON SHEET 1C SH NO. 2B

Figure 6, Sheet 3

TITLE
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202 G2,12
FUNCTIONAL ELEMENTARY





JUMPER	HYDRO EXCITATION ROTATING EXCITER	HYDRO EXCITATION BUS FED.	STATIC
F1S	1-2	1-2	2 - 3
F2S	1-2	2-3	2 - 3
F3S	2-3	1-2	1 - 2
F4S	2-3	2-3	2 - 3
POTENTIOMETER			
F1P	225	0	0
F2P	390	0	102

Figure 6, Sheet 4

REVISIONS		PRINTS TO
1	KR 5-1-80 CHANGE CHART	3EH1
2	KR 5-29-80 BH116B1BRC	4QA5
3	KR 6-30-80 CH. REF	30A1
4	KR 9-24-80 ADDED KEY KEY	30A2
5	D. Daw 8-29-82 Added LED 1&2	30A3
6	EE 2-8-84	

MADE BY W.O. SLAUGHTER ISSUED 4-6-79
APPROVALS
DSD SALEM, VA
3S7932LA202
CONT ON SHEET 2D SH NO. 2C

PRIMARY VOLTAGE	S1T,S2T TAPS TO BE USED
110	1-2
115	1-3
120	1-4
125	1-5
135	1-6

Figure 6, Sheet 5

REVISIONS		PRINTS TO	
1	MK 6-26-80 CL REV	3QA2	3CH1
2	MK 7-24-81 ADDED PART KEY	3QA3	4QA5
3	ERC 2-9-84 2-9-84		3QA1

MADE BY WD SLAUGHTER
ISSUED 4-8-79
DRIVE SYSTEMS PROD. DEPT.
SALEM, VA.
3S7932LA202

- NOT FURNISHED BY GE SALEM
- MOUNTED ON CUSTOMER CONTROL BOARD
- CUSTOMER TERMINAL BOARD
- INTERNAL TERMINAL BOARD POINT
- REDUNDANT PC BOARD PINS
- TERMINAL BOARD POINTS ON BACK OF PC BOARD RACK
- INDICATOR LIGHTS
- LIGHT EMITTING DIODE (LED)
- RELAY OR CONTACTOR COIL
- RELAY CONTACT
- RESISTOR
- POTENTIOMETER
- CAPACITOR
- INDUCTANCE, TRANSFORMER WINDING
- FUSE
- DISCONNECT SWITCH
- RHEOSTAT, ADJUSTABLE RESISTOR
- RECTIFIER (DIODE)
- THYRISTOR (SCR)
- ZENER DIODE
- OPERATIONAL AMPLIFIER (OP AMP)
- TRANSISTOR
- PUSHBUTTON (N.O.)
- PUSHBUTTON (N.C.)
- CIRCUIT COMMON
- TWISTED SHIELDED LEADS
- AND — AND CIRCUIT
- PCB MOUNTED SELECTOR SWITCH
- METAL OXIDE VARISTOR (M.O.V.)
- TWISTED LEADS

CIRCUIT	SHEET
CEILING SENSING	2B
POWER SUPPLY	
±15 VOLT	2D
24 VOLT	2C
RELAY	2C
SIGNAL CONDITIONING	2B
FREQ. TRANSDUCER	2A
WASHOUT AND OUTPUT	2B

REV NO.

REVISIONS			PRINTS TO
1	NR 6-30-80 Ch. Ref		3EH1
2	NR 7-24-80 Added PRT. ICEY		4QA5
3	EPL 2-8-84 7-24-84		3QAL
			3QA2
			3QA3

MADE BY W.D.SLAUGHTER ISSUED 4-6-79 APPROVALS
DSD SALEM, VA
3S7932LA202 SHEET 3 OF NO. 21

Figure 6, Sheet 6

GENERAL ELECTRIC

44C322516 CONT ON SHEET FL SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

50 HZ BAND PASS FILTER

FIRST MADE FOR 3S7032LA202G12

44C372627-G02

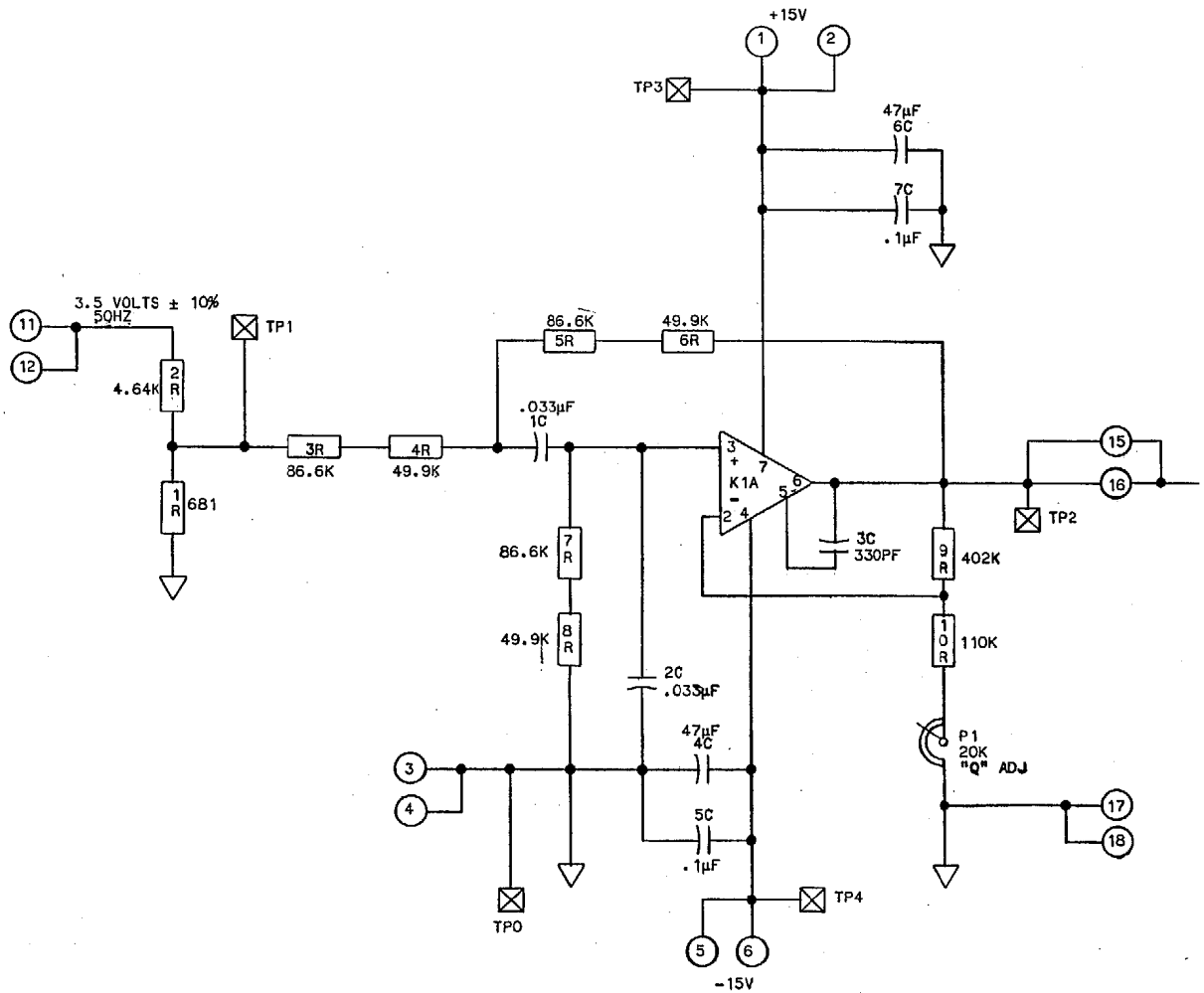


Figure 7, Sheet 1

REVISIONS

PRINTS TO 711

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C322516 CONT ON SHEET FL SH NO.

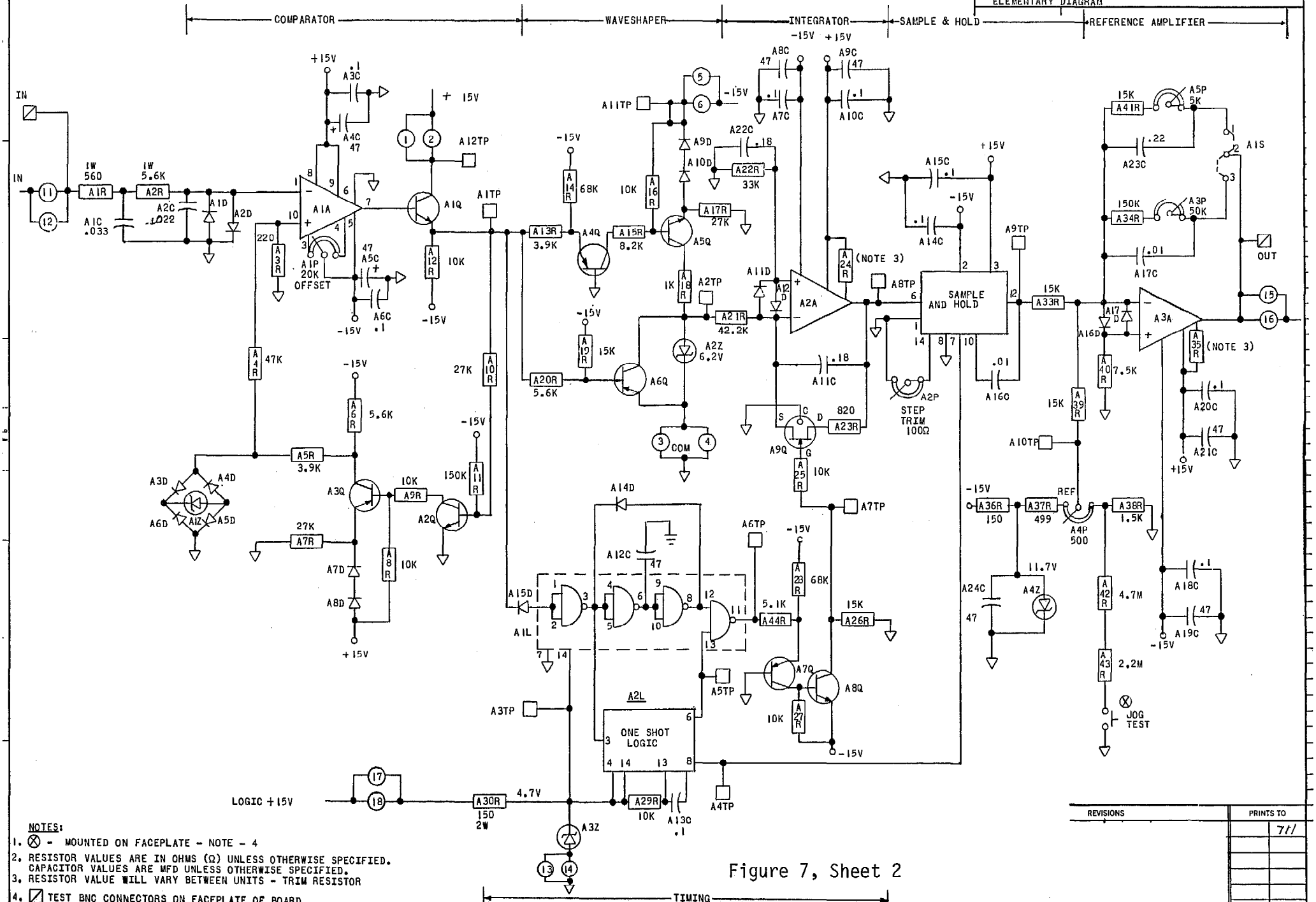
GENERAL ELECTRIC

44C322515

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
FREQUENCY TRANSDUCER
POWER SYSTEM STABILIZER
FIRST MADE FOR 44C331856-G02

ELEMENTARY DIAGRAM



NOTES:
1. ⊗ - MOUNTED ON FACEPLATE - NOTE - 4
2. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED. CAPACITOR VALUES ARE MFD UNLESS OTHERWISE SPECIFIED.
3. RESISTOR VALUE WILL VARY BETWEEN UNITS - TRIM RESISTOR
4. ☐ TEST BNC CONNECTORS ON FACEPLATE OF BOARD.

Figure 7, Sheet 2

REVISIONS | PRINTS TO 711

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C322515

GENERAL ELECTRIC 44C306563 CONT ON SHEET SH NO.

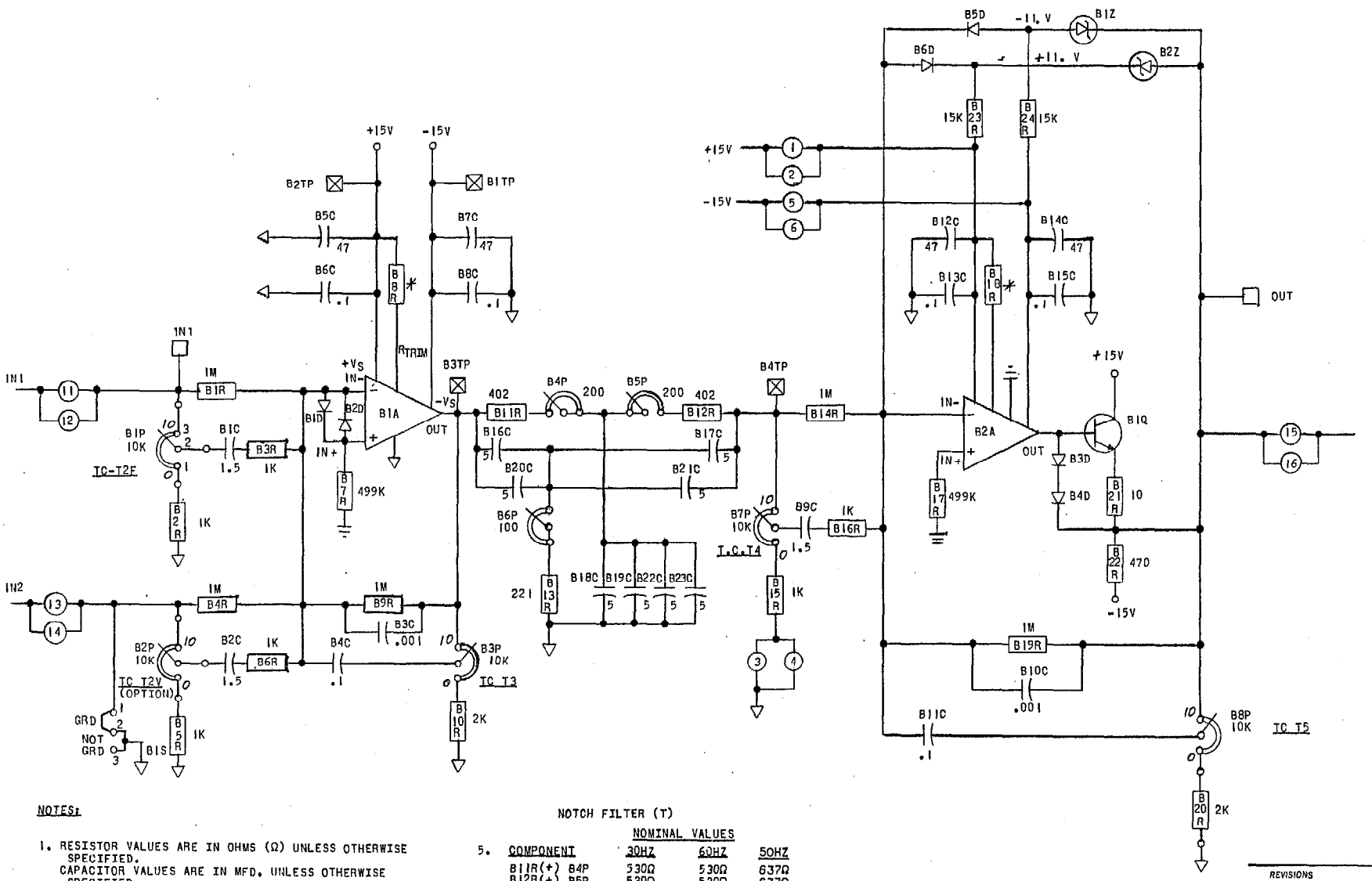
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
SIGNAL CONDITIONING
POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932MA323G1, LA202G1

ELEMENTARY DIAGRAM

I2 - I3 1ST LEAD/LAG — NOTCH FILTER — I4 - I5 2ND LEAD/LAG



NOTES:

1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
   CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
2. OMIT B20C, B21C, B22C, B23C ON 3600 RPM GENERATORS (60HZ).
3. * RESISTOR VALUE WILL CHANGE FROM ONE UNIT TO NEXT.
4. ☒ BOARD MOUNTED TEST POINT.
6. ☐ TEST BNC CONNECTORS ON FACEPLATE OF BOARD.

NOTCH FILTER (T)

5.

COMPONENT	NOMINAL VALUES 30HZ	60HZ	50HZ
B11R(+) B4P	530Ω	530Ω	637Ω
B12R(+) B5P	530Ω	530Ω	637Ω
B13R(+) B6P	265Ω	265Ω	318Ω
B16C	10MFD	5MFD	5MFD
B17C	10MFD	5MFD	5MFD
B18C, B19C, B22C, B23C	20MFD	10MFD	10MFD

Figure 7, Sheet 3

REVISIONS		PRINTS TO
4	E.G. Ellis JAN 21 1974	11
5	W.J. Shiflett KD JAN 18 1979	
6	W.J. Shiflett APR 20 1979	

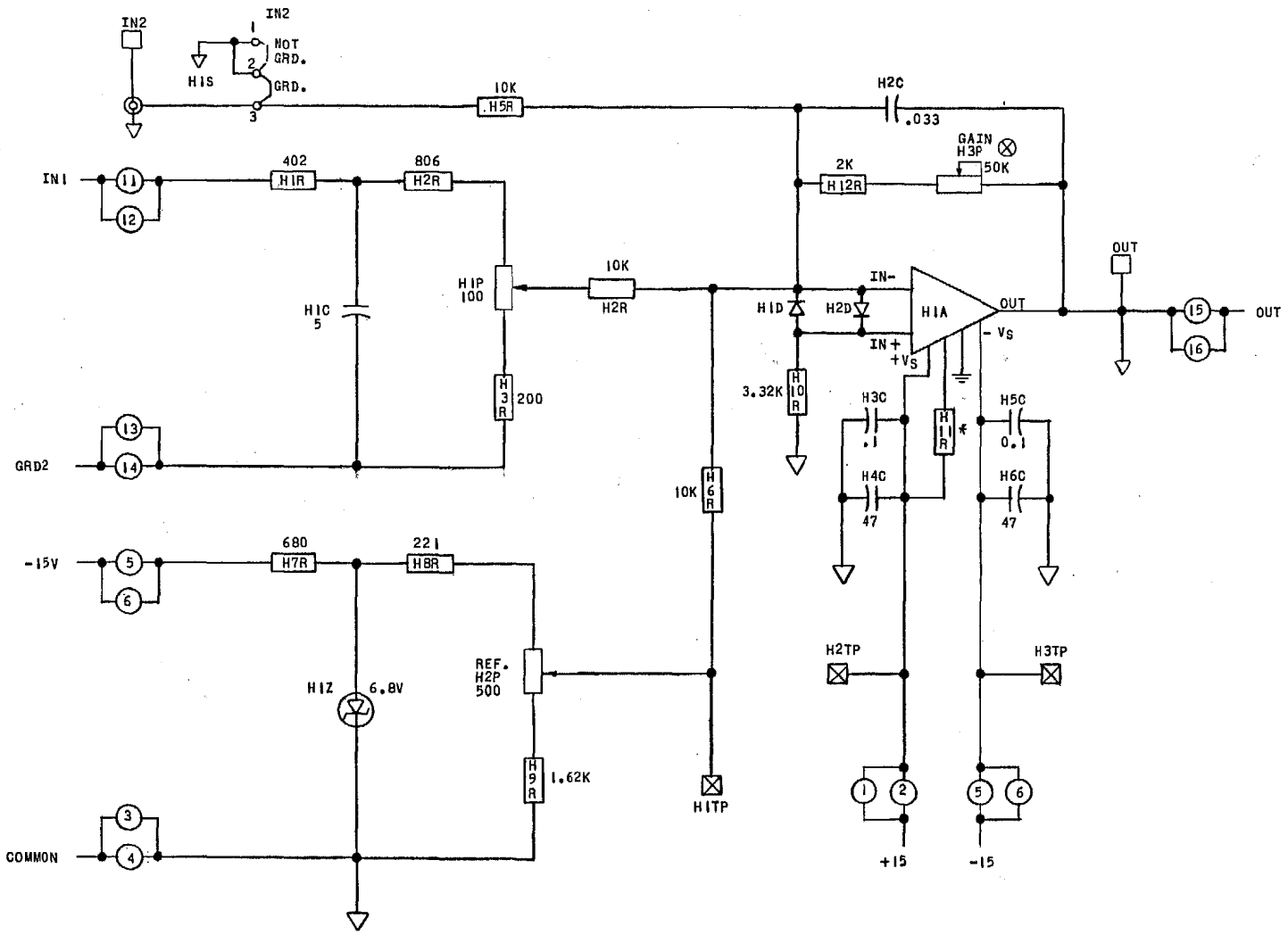
MADE BY E. Ellis 4/17/73 ISSUED R. Mayer May 18, 1973
APPROVALS T. Smith
DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44C306563 CONT ON SHEET SH NO.

DRAWING NO. 44C306563 CONT ON SHEET SH NO. REV NO.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
VOLTAGE DEVIATION SENSING
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932MA324G1, LA202G1
ELEMENTARY DIAGRAM



NOTES:

□ - MOUNTED ON FACEPLATE

RESISTOR VALUES ARE IN Ω AND ARE 1/2 WATT.
CAPACITOR VALUES ARE IN MFD.

⊗ - THROUGH FACEPLATE ADJUSTMENT

Figure 7, Sheet 8

REVISIONS		PRINTS TO
1	STANDARDIZATION OF SYMBOLS K.C.24.1 APR 23 1974	7/1

MADE BY	APPROVALS		
REV MATER 4/7/73	T. Smith	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	44C306564
D. Mayer May 18, 1973			

FP-809-P (11-72) PRINTED IN U.S.A.

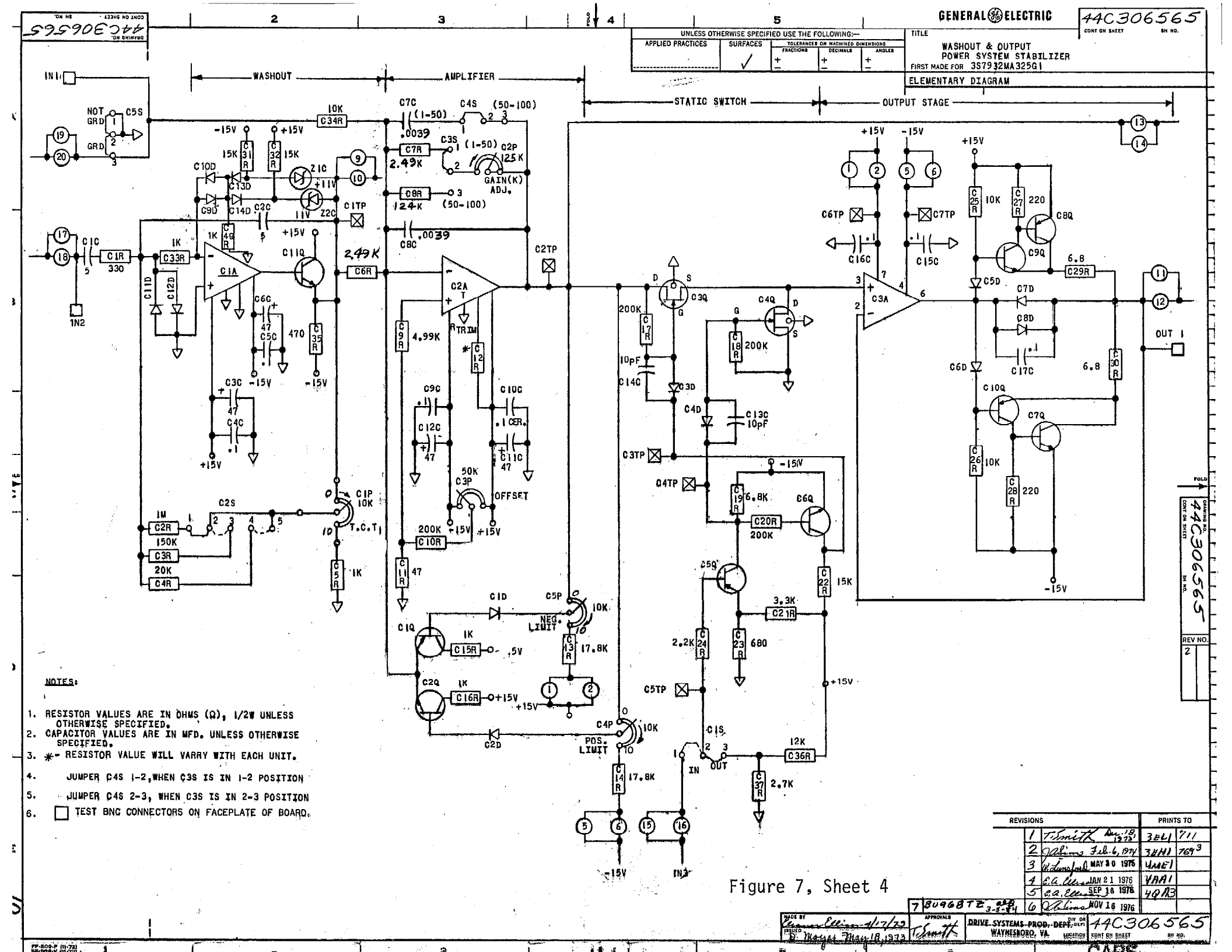


Figure 7, Sheet 4

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
UNDER-VOLTAGE DETECTION OVERVOLTAGE
FIRST MADE FOR
ELEMENTARY DIAGRAM

REFERENCE AMPLIFIER — COMPARATOR — LOGIC CIRCUIT

+15V -15V IN2 IN1 +30V GRD2 -15V 6.2V IN825 D1Z OUT1 OUT2 (TEST)
D1TP D2TP D3TP D4TP D5TP D6TP D7TP D8TP D9TP D10TP D11TP D12TP D13TP
D1A D2A D3A D1L ONE SHOT D2L D3L
INPUT VOLTAGE TRIM R_{TRIM} *
D1R 402 D2R 806 D3R 200 D4R 10K D5R 10K D6R 10K D7R 680 D8R 221 D9R 1.62K
D10R 100K D11R 10K D12R 3.9K D13R 10K D14R 3.9K D15R D16R 5.6K D17R 5.1K D18R 220 D19R 820 D20R 5.6K D21R 5.1K D22R 220 D23R 820 D24R 150Ω D25R 10K D26R 4.99K D27R 220 D28R 5.1K D29R 5.1K
D1C .1 D2C .22 D3C .1 D4C 47 D5C .1 D6C 47 D7C 2.2 D8C .1 D9C .1 D10C 22 D11C .1 D12C .1 D13C 33 D14C 220pF
D1P 100 D2P 500 D3P 2K D4P 500 D5P 2K D6P 500
D2Z 6.2V D3Z 6.2V D4Z 4.7V
NEG. LIMIT NEG. REF. POS. LIMIT POS. REF.

NOTES:
⊗ - MOUNTED ON FACEPLATE
RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
* - RESISTOR VALUE WILL VARY WITH EACH UNIT.
☒ BOARD MOUNTED TEST POINT.

Figure 7, Sheet 9

REVISIONS		PRINTS TO
1	Feb. 6, 1974	11
		769

DRAWING NO. 44C306566 REV NO. 1

MADE BY V. Klein 4/18/73 ISSUED R. Meyer May 21, 1973
APPROVALS T. Smith
DRIVE SYSTEMS PROD. DEPT WAYNESBORO, VA. LOCATION
44C306566 CONT ON SHEET SH NO.

GENERAL ELECTRIC 44C306567 CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CEILING SENSING
FIRST MADE FOR MA327
ELEMENTARY DIAGRAM

COMPARATORS — LOGIC CIRCUITS — OUTPUT STAGE

+15V
-15V
E1TP
E9C 47
E3TP
POS. LIMIT
E2P 5K
5.1K E3R
POS. REF. E4P 1K
330 E4R
330 E5R
E1Z 11.7V
E2C .1
E1R 5.6K
E1D
E2D
E1A
E1C 2.2NF
E6TP
E22Z 4.7V
E3C .1
E1L
E7TP
IN
E8TP
OUT
E3L ONE SHOT
E1Q
E7C 180
E7R 30K
E1P 500K
TIME DELAY SECONDS(S)
TIME DELAY
E1S
39.2K E8R
E9R 499K
IN RELAY SECTION
E19R 10K
E18R 220
OUT
E10TP
130 2W E6R
4.7V
E3Z
E3Q
E15R 4.3K
E16R 10K
E7D
E8D
E2TP
E10C 47
E21Z 4.7V
E2Z 11.7V
E12R 5.1K
E13R 3.3K
E2Q
+15V E14R 2.2K
E5D
E6D
E17R 27K
IN
E4C 2.2NF
NEG. LIMIT
E3P 5K
5.1K
E20R
E10R 330
E4TP -10V
E6C .1
NEG. REF. E5P 1K
E11R 330
E2L
E1P8 RESET
E8C 220PF
INPUT
E2R 5.6K
E3D
E4D
E2A
E5TP
E5C .1
E11TP

NOTES:
1. ↶ - SHOWN INCREASE CLOCKWISE ROTATION
2. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
3. CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
. ☒ TEST BNC CONNECTORS ON FACE PLATE.

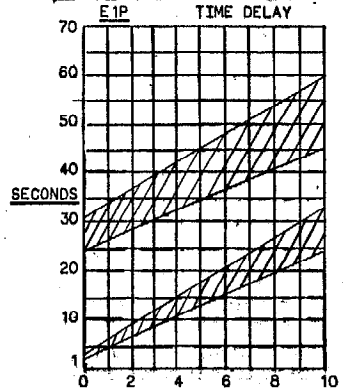


Figure 7, Sheet 6

REVISIONS			PRINTS TO
1	E.A. Clemens	Apr. 6, 1973	711
2	W. Tom	OCT 9 1 1975	768²
3	E.A. Clemens	JAN 2 1 1976	
4	D. Swick	JUL 2 1 1976	
5	E.A. Clemens	FEB 16 1978	
6	W. Tom	FEB 27 1978	

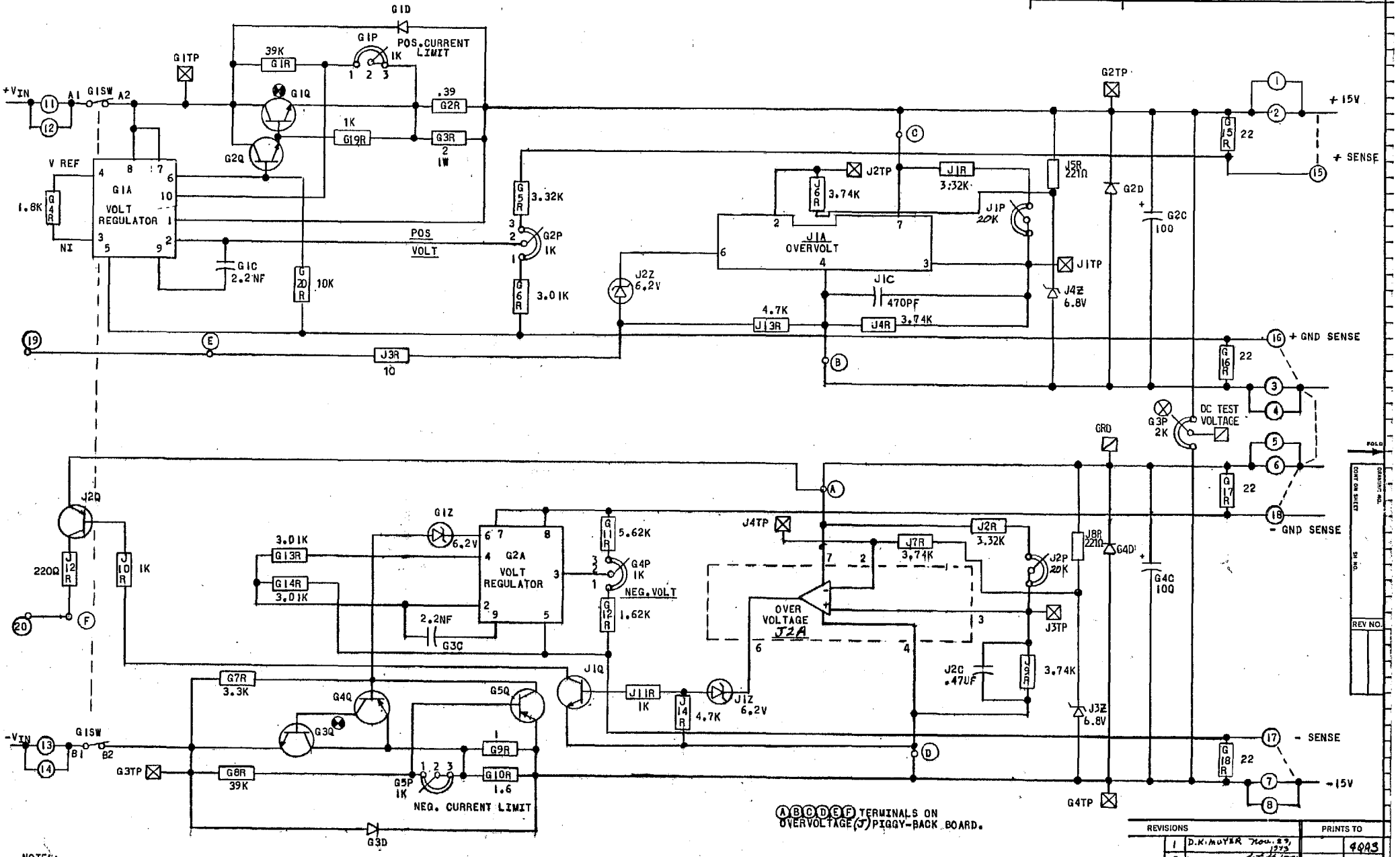
MADE BY E.A. Clemens 4/24/73 ISSUED P. Mayer May 21, 1973
APPROVALS T. Smith
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. LOCATION
44C306567 CONT. ON SHEET SH NO.

GENERAL ELECTRIC

44C306568

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
POWER SUPPLY
FIRST MADE FOR 357932MA328G1, LA202G1
ELEMENTARY DIAGRAM



NOTES:
⊗ - MOUNTED ON FACEPLATE NOTE 3, ⊗ MOUNTED ON HEATSINK
1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
CAPACITOR VALUES ARE IN MFD"S - UNLESS OTHERWISE SPECIFIED.

2. ☐ TEST BNC CONNECTOR MOUNTED ON FACEPLATE.

Figure 7, Sheet 7

REVISIONS			PRINTS TO
1	D.K. MOYER Nov. 29, 1973		4QAS
2	[illegible]		3ELI
3	[illegible] JAN 21 1976		3EHI
4	BU11638E W. LUNSFORD 4-23-79		
5	[illegible] 8-27-80		
6	BU9478M [illegible] 1/27/81		

MADE BY [illegible] 9/2/73
ISSUED D. Moyer May 18 1973
APPROVALS T. Smith

DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT.
WAYNESBORO, VA. LOCATION

44C306568
CONT ON SHEET | SH NO.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC